

SmartShelf : A Smart Library

Sumit Kumar (2024570)

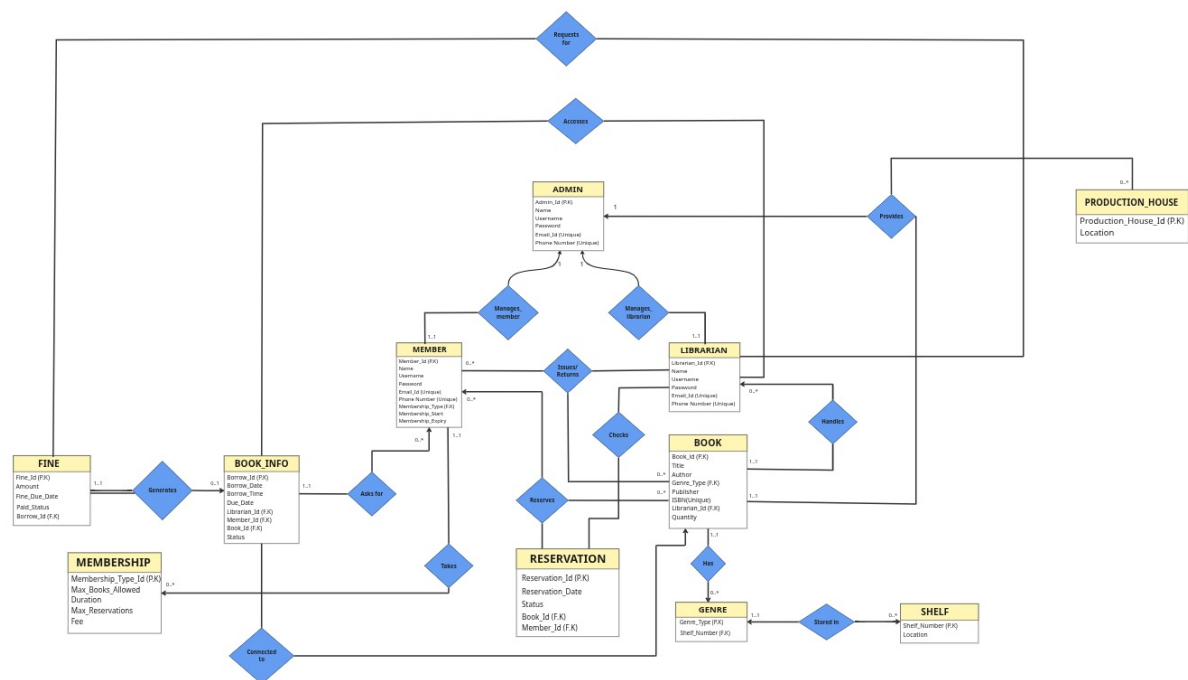
Ritanshu Sharma (2024469)

Nishant Jangid (2024394)

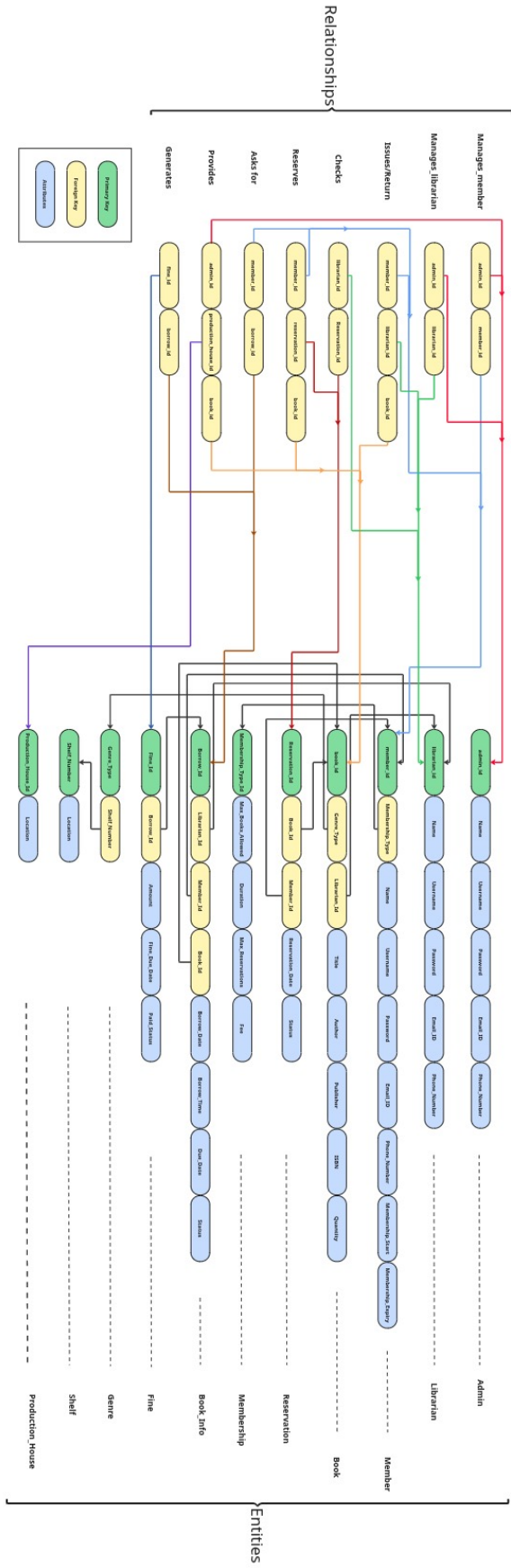
Project Group: 129 — Tutorial Group: 10

Assigned TA: Pranjal and Devyansh Chaudhary

1 Entity-Relational Model



2 Relational Model



[Link to Miro Board](#)

3 Entities

This section describes all the entities involved in the Library Management System along with their purpose and attributes.

3.1 Admin

The Admin entity represents the system administrator who oversees the library's operations. The admin manages members, librarians, and controls access to the system.

Attributes:

- Admin_ID (Primary Key)
- Name
- Username
- Password
- Email_ID (Unique)
- Phone_Number (Unique)

3.2 Member

The Member entity represents library users who can borrow, reserve, and return books according to their membership plan.

Attributes:

- Member_ID (Primary Key)
- Name
- Username
- Password
- Email_ID (Unique)
- Phone_Number (Unique)
- Membership_Type (Foreign Key)
- Membership_Start
- Membership_Expiry

3.3 Librarian

The Librarian entity represents the library staff who issue books, accept returns, and check book availability.

Attributes:

- Librarian_ID (Primary Key)
- Name
- Username
- Password
- Email_ID (Unique)
- Phone_Number (Unique)

3.4 Book

The Book entity includes all the books in the library. Each book is sorted by genre and looked after by a librarian.

Attributes:

- Book_ID (Primary Key)
- Title
- Author
- Publisher
- ISBN (Unique)
- Genre_Type (Foreign Key)
- Librarian_ID (Foreign Key)
- Quantity

3.5 Membership

The Membership entity outlines the various membership plans offered at the library, including their limitations.

Attributes:

- Membership_Type_ID (Primary Key)

- Max_Books_Allowed
- Duration
- Max_Reservations
- Fee

3.6 Reservation

The Reservation entity shows when members reserve books that are not available right away.

Attributes:

- Reservation_ID (Primary Key)
- Reservation_Date
- Status
- Book_ID (Foreign Key)
- Member_ID (Foreign Key)

3.7 Book_Info

The Book_Info entity holds details about borrowing. It includes the issue date, return date, and status of books borrowed by members.

Attributes:

- Borrow_ID
- Borrow_Date
- Borrow_Time
- Due_Date
- Return_Date
- Status
- Librarian_ID (Foreign Key)
- Member_ID (Foreign Key)
- Book_ID (Foreign Key)

3.8 Fine

The Fine entity represents penalties imposed on members for returning books late.

Attributes:

- Fine_Id (Primary Key)
- Amount
- Fine_Due_Date
- Paid_Status
- Borrow_ID (Foreign Key)

3.9 Genre

The Genre entity sorts books into different categories for easier organization.

Attributes:

- Genre_Type (Primary Key)
- Shelf_Number (Foreign Key)

3.10 Shelf

The Shelf entity refers to the physical location where books of a specific genre are kept.

Attributes:

- Shelf_Number (Primary Key)
- Location

3.11 Production_House

The Production_House entity represents publishers or production houses that supply books to the library.

Attributes:

- Production_House_ID (Primary Key)
- Location

4 Entity Relationship Roles

This section describes the roles of various entities in the Library Management System.

- **Manages:** The admin manages both members and librarians. One admin can oversee multiple members and librarians. However, each member and librarian is managed by only one admin.
- **Issues / Returns:** The librarian gives out books to members and takes care of returns. A librarian can issue several books, and each book transaction is managed by one librarian.
- **Checks:** The Librarian checks if books are available and in good condition before issuing them or accepting returns.
- **Reserves:** A member can reserve books that are currently unavailable. A member may reserve several books, and a book can be reserved by different members at various times.
- **Takes:** A member selects a membership plan. Each member is linked to one membership type, while many members can choose the same membership type.
- **Has:** A book belongs to a genre. Each book is linked to only one genre, but a genre can include many books.
- **Stored In:** Each genre is stored on a specific shelf. A shelf can hold one genre, and each genre is linked to only one shelf.
- **Handles:** A librarian takes care of books in the library. A librarian can manage multiple books, but each book is managed by one librarian.
- **Provides:** The Production House provides books to the library. A production house can supply several books, while one book can come from multiple production houses.
- **Generates:** A borrowing transaction results in a fine if the book is returned after the due date. Each fine is linked to one borrowing record.

5 Relationships

5.1 Many-to-Many Relationships

- Member and Book
- Book and Production_House

5.2 One-to-Many / Many-to-One Relationships

- Admin and Member
- Admin and Librarian
- Member and Reservation
- Librarian and Book
- Genre and Book
- Membership and Member

5.3 One-to-One Relationships

- Book_Info and Fine
- Genre and Shelf

6 Relational Schema

(1) ADMIN: admin = (Admin_Id [P.K], Name, Username, Password, Email_Id [UNIQUE], Phone Number [UNIQUE])

(2) MEMBER: member = (Member_Id [P.K], Name, Username, Password, Email_Id [UNIQUE], Phone Number [UNIQUE], Membership_Type_Id [F.K], Membership_Start, Membership_Expiry)

(3) LIBRARIAN: librarian = (Librarian_Id [P.K], Name, Username, Password, Email_Id [UNIQUE], Phone number [UNIQUE])

(4) BOOK: book = (Book_Id [P.K], Title, Author, Genre_Type [F.K], Publisher, ISBN [UNIQUE], Librarian_Id [F.K], Quantity)

(5) BOOK_INFO: book_info = (Borrow_Id [P.K], Borrow_Date, Borrow_Time, Due_Date, Librarian_Id [F.K], Member_Id [F.K], Book_Id [F.K], Status)

(6) GENRE: genre = (Genre_Type [P.K], Shelf_Number [F.K])

(7) SHELF: shelf = (Shelf_Number [P.K], Location)

(8) PRODUCTION_HOUSE: production_house = (Production_House_Id [P.K], Location)

(9) RESERVATION: reservation = (Reservation_Id [P.K], Reservation_Date, Status, Book_Id [F.K], Member_Id [F.K])

(10) MEMBERSHIP: membership = (Membership_Type_Id [P.K], Max_Book_Allowed, Duration, Max_Reservation, Fee)

(11) FINE: fine = (Fine_Id [P.K], Amount, Fine_Due_Date, Paid_Status, Borrow_Id [F.K])

P.K $\rightarrow$ Primary Key

F.K $\rightarrow$ Foreign Key

7 Contribution

With discussions and meetings, we together made up the ER Model and Relational Model for this deadline.